

USHARAL, Kazakhstan

WMO: 367290

Lat: 46.167N

Lon: 80.933E

Elev: 1302

StdP: 14.02

Time Zone: 6.00 (E06)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-8.5	-20.1	1.9	-15.0	-13.7	2.7	-8.0	28.2	15.8	22.9	18.5	2.5	70	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	94.3	66.8	91.5	66.0	88.6	64.9	69.7	88.7	68.1	86.1	66.7	84.2	4.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	92.4	76.6	61.7	86.3	75.4	60.1	81.3	75.0	34.5	89.1	33.2	86.1	32.0	84.5	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.9	18.2	13.1	DB	-22.2	99.8	6.8	2.6	-27.1	101.7	-31.1	103.2	-34.9	104.6	-39.9	106.5	
			WB	-22.5	73.4	6.7	2.0	-27.3	74.8	-31.3	76.0	-35.0	77.1	-39.9	78.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	12.2	13.6	30.0	49.1	60.9	70.5	75.2	71.9	61.3	46.4	31.7	21.3
	DBStd	23.67	10.00	10.66	11.24	8.51	7.79	6.04	4.72	5.63	7.52	8.78	10.74	9.38
	HDD50	4562	1171	1020	624	116	12	0	0	8	170	552	889	
	HDD65	7926	1636	1440	1086	479	170	19	2	153	578	1000	1354	
	CDD50	2923	0	0	4	88	350	615	781	678	347	59	1	0
	CDD65	814	0	0	0	2	43	184	318	223	42	2	0	0
	CDH74	10000	0	0	0	53	639	2050	3654	2816	761	27	0	0
	CDH80	4114	0	0	0	5	175	794	1683	1223	231	3	0	0
Wind	WSAvg	4.2	3.8	4.0	4.8	5.6	4.7	3.7	3.5	3.7	3.7	4.1	4.4	4.0
Precipitation	PrecAvg	11.2	0.9	0.7	0.8	1.3	1.0	1.0	0.9	0.5	0.4	1.0	1.4	1.3
	PrecMax	15.9	3.1	1.6	3.5	5.0	3.7	3.1	2.5	1.5	1.3	2.5	3.2	3.6
	PrecMin	5.3	0.1	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	2.6	0.6	0.4	0.7	1.1	0.9	0.7	0.7	0.4	0.3	0.7	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	38.2	66.5	80.4	88.8	95.0	98.9	97.1	90.9	77.3	59.7	44.9
		MCWB	35.6	35.8	51.1	58.1	62.0	67.6	68.7	67.2	65.4	57.5	49.7	39.4
	2%	DB	33.8	34.7	58.1	74.6	84.9	90.8	95.0	93.7	86.2	70.9	53.8	39.2
		MCWB	31.4	32.4	47.2	55.0	61.2	66.1	67.3	66.2	61.8	54.2	46.1	36.4
	5%	DB	30.0	31.4	50.8	70.3	81.2	87.9	92.1	90.7	82.0	66.7	49.7	36.2
		MCWB	27.9	29.4	42.9	53.9	60.2	64.8	66.5	65.1	59.8	52.5	43.8	34.1
	10%	DB	26.2	29.0	45.2	65.7	77.3	84.9	88.8	87.4	78.2	62.1	44.9	33.5
		MCWB	24.2	26.9	39.5	51.5	58.0	63.7	65.8	64.3	58.4	49.7	40.5	31.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	36.5	51.8	61.2	65.5	70.3	72.7	71.2	66.4	58.9	50.8	40.4
		MCDB	37.7	38.1	65.5	70.9	83.5	89.2	88.8	90.1	88.6	72.5	57.1	43.9
	2%	WB	31.9	32.7	47.3	57.1	63.2	68.1	70.3	69.0	62.9	55.3	47.3	36.9
		MCDB	33.6	34.3	56.5	71.0	80.4	86.5	88.6	88.2	82.8	69.3	53.1	38.8
	5%	WB	28.1	29.5	43.5	54.6	61.6	66.5	68.5	67.1	61.0	52.9	43.9	34.4
		MCDB	29.9	31.4	51.1	67.5	77.7	84.3	86.1	85.9	80.0	65.8	49.0	36.0
	10%	WB	24.4	27.0	39.7	52.2	59.7	64.9	67.3	65.3	58.9	50.2	40.7	31.7
		MCDB	26.1	28.9	45.1	64.5	74.6	81.7	84.1	83.2	75.9	60.8	44.7	33.4
Mean Daily Temperature Range	5% DB	MDBR	16.7	17.3	16.4	23.3	24.0	23.4	23.9	26.3	27.0	23.3	16.5	14.2
		MCDBR	18.7	16.3	24.8	31.8	31.4	27.7	29.9	31.6	33.8	30.5	23.5	16.2
		MCWBR	16.8	14.7	14.8	16.0	13.7	11.4	11.0	12.7	15.8	16.3	16.2	13.5
	5% WB	MCDBR	18.7	15.3	23.7	26.2	26.9	25.3	25.0	27.6	33.0	28.3	21.4	15.0
		MCWBR	17.2	14.0	14.9	14.4	12.5	11.6	10.7	12.5	16.9	15.5	15.4	12.8
Clear-Sky Solar Irradiance	taub	0.319	0.358	0.403	0.442	0.427	0.433	0.435	0.399	0.372	0.374	0.356	0.320	
	taud	2.069	2.042	2.072	2.045	2.149	2.197	2.210	2.289	2.314	2.203	2.143	2.089	
	Ebn at Noon	234	249	256	258	267	266	263	268	266	242	218	217	
	Edh at Noon	31	39	44	50	47	45	43	39	35	33	29	27	
All-Sky Solar Radiation	RadAvg	571	877	1241	1651	2086	2173	2086	1936	1534	956	561	438	
	RadStd	37	43	82	113	94	99	126	103	89	83	46	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page